

LSTM Based Self-Defending AI Chatbot Providing Anti-Phishing

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ABSTRACT

¹Email services have to put through a lot of effort in fighting spam emails. Most of the efforts go in for detecting and filtering spam emails from benign emails. On the other front, people are educated by banks and other organizations on the awareness of such emails. These approaches are essentially passive in nature, in countering spam attacks where the attacker is not directly engaged by the adversary. Despite all these efforts, many innocent people fall for such attacks leading them to share their account details or lose a large sum of money. We propose an AI based system, that is self-aware and self-defending, which sends coherent replies to spammers with the aim of consuming their time. To make it more difficult for spammers we reply from algorithmically generated mail servers. Also, to avoid simple match filtering of mails by spammers, we make the replies different from each other and genuine, by using a language model trained by LSTM to form sentences in natural language depending upon the context of the email.

KEYWORDS

Spam emails, Self-Defending system, Language model, deep learning, LSTM, Chatbot

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1 INTRODUCTION

One of the most common problems on the internet is email spam. It is a never-ending problem, which persists from the time of origin of mailboxes. As filtering and awareness techniques advanced over time, the percentage of spam emails also increased drastically, constituting to nearly 87% to 90% of our email traffic [2]. These emails lead to misuse of bandwidth, storage, cause loss of productivity. Along with these, spam is also one of the leading causes of legal problems like stealing private data, loss of finance, child pornography, etc. For instance, Nigerian scam emails are sent to millions of people across the world informing them that they won a lottery, which lead to a loss of \$12billion in 2013 [3].

There are many methods proposed to fight spam like, forming strong legislation against spam, modifying the existing SNMP protocol [25][26][27][28], regulating the processing of spam emails [30][31][32] and spam filters. Out of all these methods, spam filters were the most popular method adopted across email services. Manually constructed keyword patterns and blacklisting methods of known spammers were used to filter emails earlier. Later machine learning algorithms like Naive Bayesian classifiers [6], K-Nearest Neighbour [4], SVMs [5] are used by training spam emails as compared to benign emails [1].

In recent times, there has been a lot of buzz around Chatbots and its applications in real world. Big companies like Facebook and Microsoft are betting big time on these implementations. These companies have rolled out APIs (Application Programming Interface) [33] so that enterprises could use these APIs to build software ecosystem. Banks and other online e-commerce companies are also using these services to add more personal touch to their services.

In general, chatbots or dialogue systems are classified into two types; task oriented dialogue systems and non-task oriented (open domain) dialogue systems. Much of the recent research has concentrated on task-based systems [16][17][18]. In general,

these systems use center-word extraction, question-type classification, intent network, etc. to understand the intent and topic of the conversation along with dialog state to track the state of each conversation flow. Using this information, some systems try to select the most probable reply from some predefined sentences (rule-based or probability-based) [34] [35]. To address the shortcomings of rule-based systems like not scaling to unknown domains or replies being rigid; few systems use a dialogue generator along with information storage.

In this paper, we propose to attack spammers by spamming their mailbox with reply emails. We create a plugin for most of the popular browsers to forward all the emails from spam folder or manually tagged emails as spam to our central server. Once we get them, we have our spam filter and categorizing system which filters some falsely identified mails as spam, by the trusted mail servers (like Gmail, Yahoo mail, etc.). Eg: Gmail categorizes Facebook notification mails as spam. We also categorize these spams into six major classes to reply back as we use different language model for different category. We use Fasttext's word2Vec algorithm [9] to categorize these mails. We do this as each category has specific replies, which will not make much sense against other categories. The categorized emails are later sent to a *spin server* which creates new DGA domains, mostly squatted domains of different communities or organizations with interchangeable English words to make it look genuine (Eg: Table 3). Using these domains, we send reply emails to the spammers by specially trained LSTM based language model. As the model is capable of generating different domain based (e.g., financial, health, porn, etc.) word sequence similar to humans, it would be difficult for a spammer to identify a genuine victim or a bot.

Previously there have been instances to fight the spam in this way. *Blue Security Inc's* has first requested the spammers to stop sending unsolicited emails to their clients. However, when the spammers didn't mind the request, they asked their clients and employees to attack with over half a million requests to stop spamming. This could temporarily stop them, but when a counter attack was launched by a Russian spammer, the company's website and services were down [36].

This paper is organized as related work in the second section; proposed model is shown in the third section, followed by the experimental results in the fourth section.

2 RELATED WORK

Over the years, many algorithms and methods have been applied to detect and filter spam. Bag-of-words based filters take a message as input and then convert to a d dimensional vector, each labeled to either spam email or a legitimate email. One more, most common way is to share the information about spam with another user. [23] [24] does the same thing using P2P. They share the information of spammer details and reports with other P2P users in a mail server.

Chatbots have been created with a deep intention, of talking like human beings and help them with their works, like marketing, customer service and auto-replies. Some of the major challenges being faced in successful implementations of chatbot

are, not getting a proper answer, lack of context-aware, impersonal, lack of understanding real world experience.

John Bateman et al. [38] describe Natural Language Generation (NLG) as a problem of how machine can generate natural language text. Traditional NLG approaches maps all the semantic symbols to an intermediary state and then converts it to the target symbols [39][40]. In 2008 [20], Anja Belz et al. brought a combination of probabilistic and comprehensive techniques for automatic weather forecasting application. As manually tagging these corpora are very tedious, Molina et al. in [41] proposed learning sentence rules from utterance corpus labeled with Rhetorical Structure Theory (RST). This procedure is also an expensive task. To remove the manual annotation task of corpus entirely, recently the Neural network based dialog systems (RNNs and LSTMs) are introduced, which are independent on rules [41].

In case of NLG with related to chatbots, the response should be contextual. Initially this problem was mapped to statistical machine translation (SMT) problem [42]. Later [43] improved this idea by adding contextual information. In recent times Seq-To-Seq models were introduced by [43] [44] using neural networks and could successfully build end-to-end conversation systems. Researchers are using Reinforcement learning algorithm from 2010 [20], and the one of the latest work is in [21]. Learning a language from text corpus labeled with vocal words [14] is one of the common approaches in the area of NLG.

Similar to this corpora, there were interesting works based on dialogue corpora [19], which consist of proper human discussion. Gunning et al. in [15] have done a promising work on chatbot using N-gram technique, and it will extract all the content word to talk like a human being. Some of the Chatbot works are having specific tasks like technology distance education training bot [16], which are mostly task specific.

In our approach, we use Seq-To-Seq model using an stock LSTM cell on email spam and replies.

3 PROPOSED FRAMEWORK

The proposed system has five modules. The interface is a plugin through which the user can drop spam mails into a specific folder. The plugin will automatically send all these mails to our servers. We have a basic filter module to filter out falsely classified emails. After filtering the emails, each one is categorized into a pre-defined category. This categorization is done by a well-trained supervised classifier. After detecting the category, each categorized email is transferred to dedicated chatbot rooms. Each category has specially trained chatbots, and they will reply back to the spammers through the mail servers. These reply emails are from different DGA domains, which are simulated by the system. The full architectural diagram is shown in Figure 1.

3.1 Browser Plugin

We created a browser plugin in which user will have an option to create a folder in the mailbox to put the manually tagged spam mails other than the existing spam

folder. Once these tagged mails are put into the folder, they are encrypted and sent to our servers.

3.2 Data Preprocessing

This module performs the raw email data purification for making the data into the next level processing format. It will remove all redundant data, alphanumeric data and stop words. Stop word removal is performed through traditional dictionary based checking and alphanumeric data will be detected using pattern matching. Also it will perform Stemming by Porter Stemmer [15] algorithm. After the preprocessing, the next step is to filter out the spam emails using a supervised classifier.

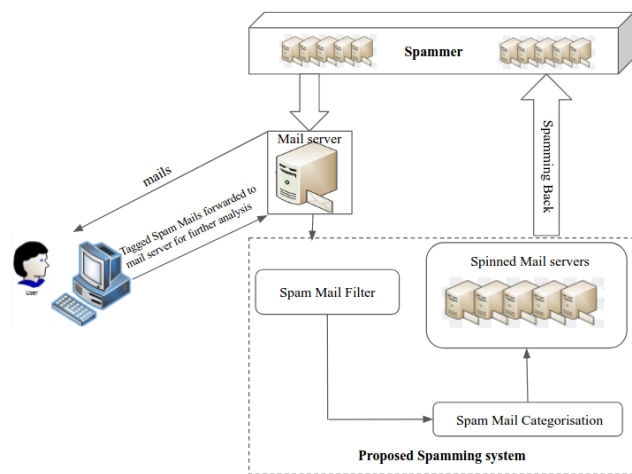


Figure 1: Architectural Diagram.

3.3 Spam Mail Filtering

After preprocessing the emails, the next step is to filter the spam emails. Initially we filter all the known domains (like facebook mail, twitter, etc.) if any, as sending a reply to them does not make much sense. Also, we filter out emails having 'noreply' keyword in its subject or content as most of the replies to these mails are dropped. Once these pre-filterings are done, we send to a classification algorithm, in our case Random Forest to detect additional spam mails. The features that are taken into consideration are a bag of word model after the pre-processing. Also since we have an aggregated view of these emails from different users, we can prioritize based on the frequency of occurrence and it will become an added feature for our training. We have tested the dataset with different algorithms and finally concluded to use Random forest based on its accuracy.

3.4 Spam Mail Categorization

In general, we get spam emails from different sources like finance companies, job search portals, line of software products etc. To spam the spammers, we should have a clear idea of the category, which the received mail belongs to. Spam emails categorizations, which we considered are:

1. Finance: Mails related to lucky winner, bank related offers
2. Computer related: All the emails, which are related to software like mobile phones, software etc. are, comes in this category.
3. Health related: All the emails related to diet/weight loss etc. will come in this category.
4. Job-Related: Dream job-related spam emails are categorized in this class.
5. Porn Spam
6. Other Spam

Some of the words which are used by the spammers are '\$,' 'offer,' 'congratulations', 'click,' 'free,' 'awarded,' 'you won,' etc.

We have analyzed different publicly available datasets, found out the uniqueness/features in each category, and trained. Collected emails are trained using the advanced version of word2vec algorithm with the help of Fasttext[9], an extremely fast and robust software for text classification. This step is inevitable to train our chat bot in a sophisticated way. We have collected data from enron [10], Ling-Spam dataset[11], spambase[12] to train the spam emails for categorisation. From our study, we could identify that many of the spammers are using disposable mail ids to get rid of domain blockers. We have found the following mail which is shown in Figure 2 the mail id 's.srinivasan <info@2panel.com>' which is created using the website: Wegwerf-eMail-Adresse-Kostenlose provisorische E-Mail-Adressen [8].

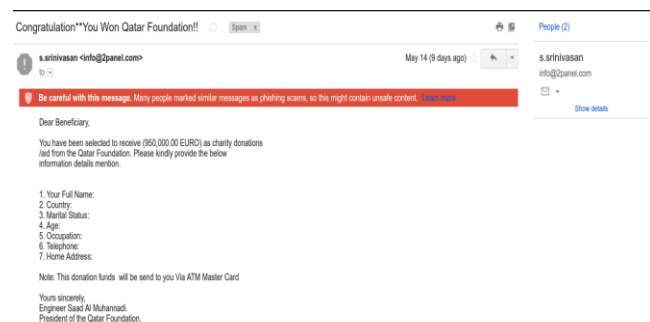


Figure 2: Sample Spam Mail.

3.5 Spin Server

Most of the time spammers send a lot of spam emails across the globe. In the case of a large volume spam, typically they send to millions of email ids. In general, the source of these email ids is from forums, blogs, social networking sites like Twitter, other e-commerce sites, etc. Because their target is not specific, their list of email ids consists of various domains, organizations, etc. Since it is tedious for the spammers to track all the email domains they sent to, we can send replies from random domains created using Domain Generated Algorithms (DGA) to these spammers.

DGAs are used to create domains randomly on a large scale for registration. We use English words and domain squatted word (of organizations, universities, places, communities, etc.) while generating random domains. Squatted words are almost similar to legitimate words except for small changes in the name. (Eg: 'amrita' and 'amrta' or 'google' and 'g00gle' looks similar but they are actually different). Registering such names as domains and attacking is a popular way for botnet creation. Furthermore, usage of alternate Top Level Domain (TLD) is another type of squatting mechanism around the world. Eg: www.keralapsc.gov.in is an Indian website whereas ww15.keralapsc.org is a Chinese website. Techniques we used for similar domain name generation are Fat Finger Distance, Damerau-Levenshtein[37]. That being said, in our paper we only use these techniques for confusing spammers. By this way, it would be hard for the spammers to find out the spam-reply-emails (SRE) from legitimate-reply-emails (LRE).

The *spin server* module takes care of creating such domains at a fixed period. We keep each DGA domain for a month before refreshing it. After creating a domain, a name server is created along with a mail server automatically. We create around thousand usernames for each domain. These usernames are created from a named dataset consisting of one million names collected from various websites across ten different countries including few African countries. We keep these mail servers for a month as we expect a response from the spammers, which can lead to a conversation causing them to invest more time on our bot.

3.6 Language Model

The most important module in our framework is the language model as it interacts with the spammer himself/herself. The more convincing the model generates sentences, more spammer's time is wasted. We use the word *seq2seq model* proposed in [43] where given an input

sentence $S = \{w_1, w_2, w_3, \dots, w_n\}$ each LSTM cell at a particular time t takes input a 1-hot encoding $\mathbf{e}_t$ of token w_t using the input gate i_t , memory gate f_t and outputs a vector representing the probability distribution of the next token $\mathbf{e}_{t+1}$ using an output gate o_t . $\mathbf{C}_t$ is called the cell state vector, and $\mathcal{J}$ is the sigmoid function, which acts as a gate to the cell. The model generates one token per time step till a stop signal is generated or the condition is met. The proposed LSTM architectural diagram is shown in Figure 3.

Since LSTMs are famous and are widely adopted in many fields, people have come up with a variety of architectures. Out of them, the basic cell proposed by (Hochreiter and Schmidhuber, 1997) is most popular, and that is what we will use in our model. These are the following equations related to it

$$i_t = \mathcal{J}(\mathbf{W}_{wi} \mathbf{e}_t + \mathbf{W}_{hi} \mathbf{h}_{t-1}) \quad (1)$$

$$f_t = \mathcal{J}(\mathbf{W}_{wf} \mathbf{e}_t + \mathbf{W}_{hf} \mathbf{h}_{t-1}) \quad (2)$$

$$o_t = \mathcal{J}(\mathbf{W}_{wo} \mathbf{e}_t + \mathbf{W}_{ho} \mathbf{h}_{t-1}) \quad (3)$$

$$\mathbf{C}_t = \tanh(\mathbf{W}_{wc} \mathbf{e}_t + \mathbf{W}_{hc} \mathbf{h}_{t-1}) \quad (4)$$

$$\mathbf{C}_t = f_t \cdot \mathbf{C}_{t-1} + i_t \cdot \mathbf{C}_t \quad (5)$$

$$\mathbf{h}_t = o_t \cdot \tanh(\mathbf{C}_t) \quad (6)$$

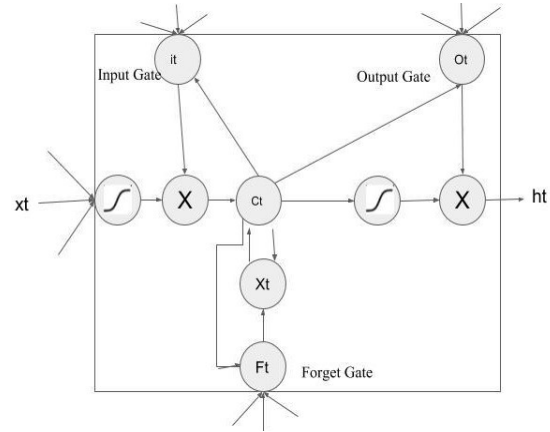


Figure 3: **Proposed Architectural Diagram using LSTM** [14].

To make the model understand the sentence formation in English language and narrow down the probable words (e.g.: In the English language, 'in' 'is' are mostly propositions and it is rare to see propositions as any other parts-of-speech.) we send POS tags also along with the word. We convert the POS tag of a token at time t to a 1-hot vector $\mathbf{p}_t$ and then append it to the 1-hot vector of the token itself $\mathbf{e}_t$. Now the 2-hot vector is sent into LSTM. Now the equations change to

$$i_t = \mathcal{J}(\mathbf{W}_{wi} [\mathbf{e}_t, \mathbf{p}_t] + \mathbf{W}_{hi} \mathbf{h}_{t-1}) \quad (1)$$

$$f_t = \mathcal{J}(\mathbf{W}_{wf} [\mathbf{e}_t, \mathbf{p}_t] + \mathbf{W}_{hf} \mathbf{h}_{t-1}) \quad (2)$$

$$o_t = s(\mathbf{W}_{wo} [\mathbf{e}_t, \mathbf{p}_t] + \mathbf{W}_{ho} \mathbf{h}_{t-1}) \quad (3)$$

$$\mathbf{C}_t = \tanh(\mathbf{W}_{wc} [\mathbf{e}_t, \mathbf{p}_t] + \mathbf{W}_{hc} \mathbf{h}_{t-1}) \quad (4)$$

3.7 Datasets

We manually collected reply message dataset for all the categories we have categorized above. We collected from social media like Twitter and product forums. For the categories which we could not collect (like porn spam), we manually generated replies. We use a question generator to add more topic related conversation. This could lead the spammers to reply to it making them invest time in our email. In addition, since we send these emails in bulk from many DGA mail servers, it would be very difficult for the spammers to track the genuine victim emails.

4 EXPERIMENTS AND RESULTS

As you can see our system has mainly five modules: browser plugin, spam email filtering, categorization, spinning the domain and reply back to the spammer. We have collected email data from different publicly available dataset for training. The collected dataset is trained on a Random Forest, Decision tree, Multi-layer Perceptron, Support Vector Machine and Naive Bayes. Out of them Random Forest outperformed with an accuracy of 95.414% on the same and the details are shown in Table 1. The ROC curve of a Random Forest is shown in Figure 4.

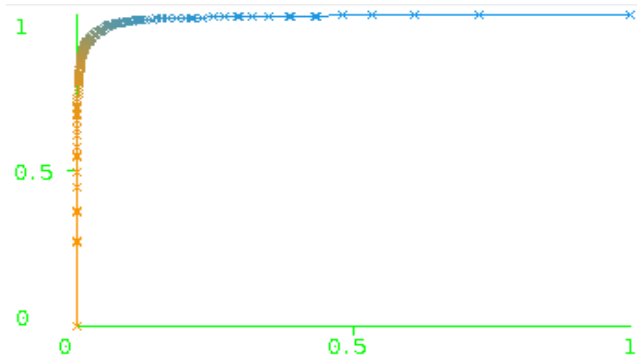


Figure 4: ROC Curve of Random Forest.

Table 1: Accuracy Details

Classifier	Accuracy
Random Forest	95.414%
Decision Tree	92.9798%
Multi-Layer Perceptron	91.0454%
Support Vector Machine	83.6557%
Naïve Bayes	79.2871%

For categorizing the emails, we have used neural network using word2vec algorithm. This algorithm will take the whole sentence and make a model according to the word sequence. It will consider the closeness of neighbouring words. It is a hierarchical classifier instead of a flat structure, in which the categories are organized as a tree. We have used the fasttext for this purpose and the parameters we have used are explained in the table 2. Dataset has 36,000 unique emails, which are equally distributed among six categories. Each category was trained by 5000 emails and the cross validation and testing are done with 500 each. Using this data, the system could achieve 90.5% accuracy. The system has a module, which continuously retrain the model for a particular interval to increase the accuracy.

Table 2: Classifier Parameters

Lr: 0.5	<i>bucket : 2000000</i>
<i>lrUpdateRate: 100</i>	<i>minn : 3</i>
dim : 100	<i>maxn : 6</i>
Ws : 2	<i>thread : 1</i>
Epoch: 1000	<i>t : 1e-4</i>
minCount: 5	<i>label : __label__</i>
<i>minCountLabel : 0</i>	<i>verbose : 2</i>
<i>Neg: 5</i>	<i>pretrainedVectors :</i>
<i>wordNgrams: 2</i>	<i>Pretrained word vectors</i>
<i>loss : ns</i>	<i>for supervised learning</i>

For spinning the domains, we have employed fat finger algorithm and our English dictionary based algorithm. After getting the domains, an LSTM based dedicated chat bots will be replying back to the spammers according to their category.

Use Cases with our language model:

Our system could successfully fool some of the spammers. Some of the conversations went on till 19 conversations before the spammers could give up or understand that they were going nowhere with the conversation. Following is one instance:

We got a spam email from s.srinivasan <info@2panel.com>

As you can see that (in Table 4), the replies from the bot are decent if not good. The LSTM model makes some mistakes while forming sentences like in the case of the last conversation (*What is the requirement do it now?*) where it does not make much sense. Also in the sentence *Who Your Full Name do you want?*, we can see that it filled the same variable detail directly.

Table 3: Sample Spinner Mail Ids

chandnixhg@amrta.online	xephania@itt.ed
abimbola@amrta.la	aatma@iitu.com
abioye@amrti.la	kanhaiya@iitu.com
dhandnixhhg@amrti.onlne	kamaljeet@iitu.com
abdurahman@iitu.com	tadala@amti.in
saatvak@amrti.onln	abdulaziz@iitu.com
dikeledi@amrti.onln	vijay@iitu.com
ebele@amrti.la	vikrant@iitu.com
kwanza@amrti.la	virendar@iitu.com
nomusa@amrti.la	vishwa@iitu.com
nontle@amrti.la	udai@iitu.com
surendran@amrti.la	upendra@iitu.com
olamilikan@gmail.la	utpal@iitu.com
solomon@anrti.com	uttam@iitu.com
dhandni@amrti.in	shambhu@iitu.com
yewande@amrt.online	siddharth@iitu.com
zephania@amrti.in	somnath@iitu.com
abdulgani@iitu.com	subira@amti.in

Table 4: Sample Chatbot Reply to the Spammer

Spammer:	Dear Beneficiary, You have been selected to receive (950,000.00 EURO) as charity donations aid from the Qatar Foundation. Please kindly provide the below information details mention. 1. Your Full Name: 2. Country: 3. Marital Status: 4. Age: 5. Occupation: 6. Telephone: 7. Home Address: Note: This donation funds will be sent to you Via ATM Master Card Yours sincerely, Engineer Saad Al Muhannadi. President of the Qatar Foundation.	
Bot:	Hi Engineer Saad Al Muhannadi, I am really happy about it! What should I do?	
Spammer:	Hello Saatvik, You have been selected to receive (950,000.00 EURO) as charity donations aid from the Qatar Foundation. Please kindly provide the below information details mention. 1. Your Full Name: 2. Country: 3. Marital Status: 4. Age: 5. Occupation: 6. Telephone: 7. Home Address: Note: This donation funds will be sent to you Via ATM Master Card Yours sincerely, Engineer Saad Al Muhannadi. President of the Qatar Foundation.	
Bot:	I am really happy about it! What is the requirement do it now?	
Spammer:	Hello Saatvik, We would forward your details to our accounts department, and they would transfer the amount to you. It should take at most two days. Yours sincerely, Engineer Saad Al Muhannadi. President of the Qatar Foundation.	
Bot:	What do you now understand? When can you transfer the charity amount?	
Spammer:	Hello Saatvik, I don't understand. We have picked you for our charity program and would like to know your details so that we could transfer you the charity amount. Please kindly provide the below information details mention. 1. Your Full Name: 2. Country: 3. Marital Status: 4. Age: 5. Occupation: 6. Telephone: 7. Home Address: Yours sincerely, Engineer Saad Al Muhannadi. President of the Qatar Foundation.	
Bot:	I am really happy about it! Who Your Full Name do you want?	

This we have rectified later. We are still working on it to improve it further. A much larger dataset would help build better language models.

We have currently tested our framework by creating only few DGA domains. However, we can scale this to any number as the process is automated. During our two-month testing period, we sent around 11,405 replies. The distribution of the replies is shown in Table 5.

Table 5: Distribution of Replies

Category	Percentage of replies
Job Search	35%
Finance	34%
Health	20%
Computer	8%
Porn and other	3%

Around 4% of our mails received replies from spammers. Out of them 129 conversations were more than five messages long. We observed that many spammers stopped replying after first or second reply, if the generated reply is not related or vague from bot. The longest conversation with a spammer was of 17 messages. Nonetheless, our aim to make them spend their time on a bot was successful.

5 CONCLUSIONS

The proposed system can filter the spam emails and categorized into six major classes. The system can predict the category of the email, and accordingly it will reply back to the spammer's email. It spins different domains for the email using Domain Generation Algorithm and hence avoids filter of domains by the spammer. We used LSTM based language model along with question generations techniques to continue the conversation.

We believe that, the more we spam these spammers, it would make them hard to track the victims. Also as the reply-spam increases, in the long term there would be a decrease in the trend of using spam for stealing, propagation, etc.

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